

Yelp Sentiment Analysis: Elevating Customer Experience & Fueling Business Growth

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Abstract

Aiming to empower consumers and offer valuable insights to businesses for continuous improvement, this project proposes a solution for Yelp's rating system by incorporating time series analysis and sentiment analysis, providing dynamic visualizations and insights beyond static star ratings.

Introduction

Problem: Yelp's current star rating system oversimplifies business evaluations, providing an aggregated score that may not reflect recent changes in quality. Businesses lack detailed feedback for targeted improvement.

Importance: Businesses need specific insights for growth. Our solution incorporates sentiment analysis over both document and aspect level sentiments and time series forecasting to offer dynamic insights for informed decision-making and improved customer experiences.

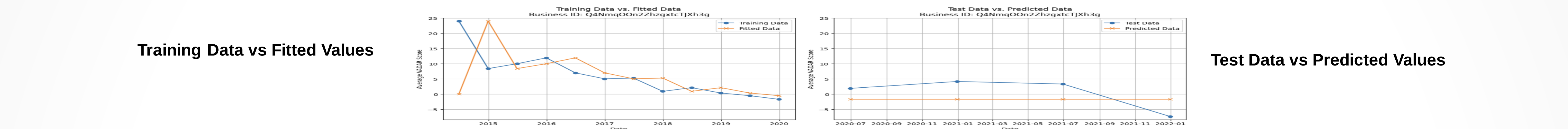
Data and Methods

Data Acquisition: We obtained our data through Yelp Open Dataset, which contains both business-related information and customer reviews. The dataset was accessed and merged using AWS Athena, which were later processed through multiple tools such as OpenRefine and common Python libraries.

	Size on Disk	Business #	Total Record #	Sentiment Score Range	Date Range
Pennsylvania	1.20 GB	31,638	1.59*1e6	[-0.99, 0.99]	2005/2 - 2022/1
Florida	0.82 GB	24,017	1.16*1e6	[-0.99, 0.99]	2005/3 - 2022/2
California	0.25 GB	5,203	3.45*1e5	[-0.99, 0.99]	2005/3 - 2022/1

Sentiment Analysis Approaches: We initially used AFINN sentiment analysis and later transitioned to VADER scoring for improved accuracy and increased nuance in sentiment scoring. Sentiment scores were aggregated into half-year intervals to address data sparsity for a more reliable analysis.

Time Series Forecasting: Our time series forecasting employs the Auto Regressive Integrated Moving Average (ARIMA) model. The dataset was split into training and test sets chronologically, optimizing hyperparameters to achieve accurate predictions. Mean Squared Error (MSE) served as our evaluation metric.



Innovation and Effectiveness:

The innovation is a combination of sentiment analysis (document/aspect level, NER generation for reviews), time series forecasting, and polarity word clouds.

Time series forecasting highlights perceived customer sentiment toward a business weighted by recency of reviews and provides a predicted sentiment rating for the next 6-month period.

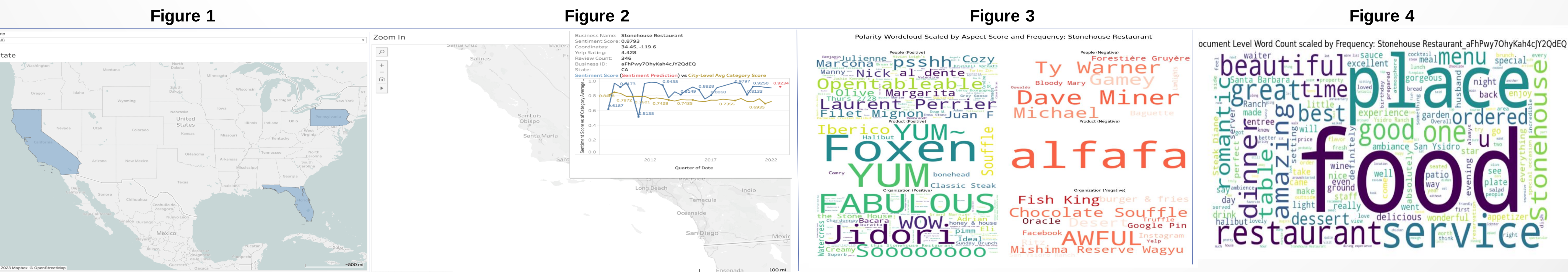
- This gives a more accurate picture of how the business has been trending and gives the customers more information than a simple mean rating.

Polarity word clouds will help business owners easily identify areas of improvement compared to just searching through customer reviews on their own. Polarity word clouds are not as common as regular word clouds in sentiment analysis, and ours are generated using the Vader score of each sentence.

- After identifying the main object (NER) within the sentences to get positive and negative word clouds at each aspect level, business owners get a summative picture on how customers view their business.

Experiments and Results

Approach	Evaluation	Results	Comparison																
OpenRefine	Manual judgment of city names/clusters	No city duplicates, casing uniform, clustered correctly	Geocoding – too computationally intensive																
Vader Scoring	N/A	Figures 3, 4	AFINN – simple to use but lacked nuance, generated scores were not standardized to any scale																
ARIMA Forecasting (6-month periods)	Mean Squared Error	<table><tr><td></td><td>Total Business #</td><td>Forecasted Business #</td><td>Test MSE</td></tr><tr><td>Pennsylvania</td><td>31,638</td><td>29,237</td><td>0.22</td></tr><tr><td>Florida</td><td>24,017</td><td>21,704</td><td>0.23</td></tr><tr><td>California</td><td>5,203</td><td>4,801</td><td>0.17</td></tr></table>		Total Business #	Forecasted Business #	Test MSE	Pennsylvania	31,638	29,237	0.22	Florida	24,017	21,704	0.23	California	5,203	4,801	0.17	- Attempted predictions of "next n reviews" – data too sparse, inaccurate and ineffective insights for businesses - Tried aggregating over fewer months – review data for many businesses too sparse, had to choose at least 6 months
	Total Business #	Forecasted Business #	Test MSE																
Pennsylvania	31,638	29,237	0.22																
Florida	24,017	21,704	0.23																
California	5,203	4,801	0.17																
Tableau + Flask	User Testing	Figures 1, 2	TabPy - Complex setup process, unsatisfactory computation time, limited functionality and accessibility, unavailable on Tableau Public																



Conclusions

Project holds significant potential in two dimensions:

1. Consumer empowerment: offering a nuanced comprehension of businesses and transcending the constraints imposed by static star ratings empowers consumers with information
2. Business improvement: offering businesses valuable insights through sentiment analysis and time series forecasting enables them to make informed enhancements

Limitations

1. Server setup requirement
2. Limited computation resources

Future Direction

1. Geographical Expansion for all the states in U.S.
2. Implement real-time sentiment analysis to capture the most current consumer experiences
3. Analysis Expansion by adding additional analysis and visualization into Flask server
4. Integrate new tools for more intuitive aspects sentiments analysis of businesses
5. Setting up sever over cloud instead of current simple local server.
6. Provide end-user testing by providing the service to interested business

Acknowledgments

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